

Topic 2: How Large Language Models Work

2.1 What is an LLM?

- A Large Language Model (LLM) is a neural network trained on massive amounts of text to predict the next token (word piece) given previous tokens
- Training data: books, websites, code, research papers -- billions to trillions of tokens
- The model learns statistical patterns in language -- which words follow which in which contexts
- Popular LLMs: GPT-4, Claude, Gemini, LLaMA, Mistral
- LLMs do not 'know' facts the way humans do -- they predict plausible text based on patterns

2.2 Tokens and Context Windows

- Token: the basic unit an LLM processes -- roughly 3/4 of a word on average
- 'Hello, world!' is about 4 tokens; a typical paragraph is 60-100 tokens
- Context window: the maximum number of tokens the model can 'see' at once (prompt + response)
- Modern models: GPT-4 Turbo (128k tokens), Claude 3 (200k tokens)
- Exceeding the context window causes the model to 'forget' earlier parts of the conversation
- Everything in the context window has equal visibility to the model -- order still matters

2.3 Temperature and Sampling

- Temperature controls randomness: 0 = deterministic (always most likely token), 1 = creative, 2 = chaotic
- Low temperature (0.0-0.3): best for factual tasks, code, data extraction -- consistent outputs
- High temperature (0.7-1.0): best for creative writing, brainstorming -- varied outputs
- Top-p (nucleus sampling): model samples from the smallest set of tokens whose cumulative probability exceeds p
- For most prompt engineering tasks, temperature 0 or 0.2 gives the most reliable results

2.4 How the Model Reads Your Prompt

- The model processes your entire prompt as a sequence of tokens and attends to all of it simultaneously
 - It does not 'plan ahead' -- it generates one token at a time, left to right
 - Instructions placed earlier in the prompt generally have stronger influence
 - The model has no memory between separate conversations -- each call starts fresh
 - System prompt (if supported): a special instruction block that sets the model's behaviour for the entire conversation
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